The Saturday Mornings.tv

TODAY'S FORECAST
BRIGHT SPELLS,
DEVELOPER MODE BY NOON

THE FREE EDITION
SAMSUNG TV MANUAL
BACK TO THE FRONT PAGE

DO-IT-YOURSELF SUPPLEMENT · PART TWO SATURDAYMORNINGS.TV

"Please adjust your set"

THE FREE EDITION · SAMSUNG LANE

SAMSUNG TV, STEP BY STEP

How to put the Saturday Mornings TV app on your own Samsung Smart TV: Samsung's tools, a certificate with your name on it, a secret handshake with the remote, one line of settings and one script.

14 STEPS | 1 SAMSUNG ACCOUNT | 1 REMOTE CONTROL | TIZEN TVS

Difficulty, in bowls of cereal: 🥣 easy · 🥣🥣 fiddly · 🥣🥣🥣 pour yourself another bowl

FIRST: BUILD YOUR STATION ANDROID TV GUIDE THE SHORT VERSION FRONT PAGE

WHAT YOU'LL NEED

- [ ] **A working station**: its address answers `{"ok":true}` at `/api/health`. Not yet? Build it first.
- [ ] **A Samsung Smart TV** (they run Samsung's Tizen system) and its remote.
- [ ] **A computer** with a terminal. The build script is written for Mac and Linux.
- [ ] **The same Wi-Fi** for the TV and the computer. The computer talks to the TV directly.
- [ ] **A Samsung account**, free. Samsung TVs only install apps signed by someone with one.
- [ ] **Tizen Studio**, Samsung's free developer tools (several gigabytes).

- [ ] **The code** from github.com/RonzGore/saturday-mornings, already on your computer from the station guide.

TIME, FIRST GO	MOSTLY SPENT	COST	EACH UPDATE AFTER
about 2 hours	downloading	$0	5 minutes

PART 1 · THE TOOLS *Samsung's toolkit on your computer*

1

Install Tizen Studio

Tizen is the system inside Samsung TVs; Tizen Studio is Samsung's kit for building apps for it. Download it from Samsung's developer site (`developer.samsung.com`, under Smart TV › Tizen Studio) and install it into your home folder, as `tizen-studio`. Our script looks for it exactly there: `~/tizen-studio`.

TYPE THIS TO CHECK

```
~/tizen-studio/tools/sdb version
```

> **YOU SHOULD SEE:** a version number. (`sdb` is the tool that talks to the TV.)

30 MIN

2

Add the TV extensions

Out of the box, Tizen Studio knows about watches and phones, not TVs. Open the **Package Manager** (it opens after installing, or from Tizen Studio's Tools menu), go to the **Extension SDK** tab and install **TV Extensions** and the **Samsung Certificate Extension** (you'll need it in step 8).

> **YOU SHOULD SEE:** both marked as installed in the Package Manager. Make a sandwich while it downloads.

20 MIN

3

Open a Samsung account

Free, at `account.samsung.com`. It's what makes your certificate official: Samsung TVs only install apps signed with a certificate from a Samsung account.

> **YOU SHOULD SEE:** you can sign in at `account.samsung.com` with your email and password.

5 MIN

4

Find your computer's address

Your TV will only accept apps from one computer, yours. It needs your computer's IP address (its number on your home network). On a Mac: **System Settings** › **Wi-Fi** › **Details**. Or ask the terminal:

TYPE THIS (MAC)

```
ipconfig getifaddr en0
```

> **YOU SHOULD SEE:** something like `192.168.1.20`. Write it down. (On Linux, `hostname -I` shows it.)

2 MIN

PART 2 · THE TV *The secret handshake*

5

REALLY!

Turn on Developer Mode

Developer mode lets the TV accept an app from your computer instead of the app store. Pick up the remote:

1. Open **Apps** on the TV.
2. Press 1 2 3 4 5 on the remote's number keys (or the on-screen number pad, on remotes without numbers).
3. Switch **Developer mode** to **On**.
4. Type your computer's IP address from step 4, and press OK.
5. Restart the TV: hold the power button until it switches off and back on.

> **YOU SHOULD SEE:** a developer mode note on the Apps screen after the restart. Yes, really. That's the whole handshake.

10 MIN

6

Find the TV's address

Now the other way round. Your computer needs the TV's IP address. On the TV: **Settings › General › Network › Network Status**, then **IP Settings** (on newer models, Network sits under **Connection**).

> **YOU SHOULD SEE:** an address like `192.168.1.50`. Write it down. Every command below uses it; swap in your own numbers.

3 MIN

7

Shake hands with the TV

With the TV on, connect to it from your computer. Same Wi-Fi, or it won't even wave back.

TYPE THIS (YOUR TV'S ADDRESS)

```
~/tizen-studio/tools/sdb connect 192.168.1.50
~/tizen-studio/tools/sdb devices
```

> **YOU SHOULD SEE:** "connected to 192.168.1.50:26101", then the TV listed as a `device` with its model name.

5 MIN

PART 3 · THE CERTIFICATE *Your name on the app*

8

TRICKY!

Make your own Samsung certificate

A certificate is a digital signature that says "I made this app, and this TV may run it". It comes in two halves, an **author** certificate (you) and a **distributor** certificate (Samsung's permission, tied to your TV's ID number, called the **DUID**). Make your own; never use anyone else's.

1. Keep the TV connected (step 7). In Tizen Studio, open **Tools › Certificate Manager** and press +.
2. Choose **Samsung**, then device type **TV**.
3. Name the profile anything you like, say **MyFamilyTV**. You'll tell the build script this name in step 11.
4. **Author certificate:** create a new one. Type your name and a password you will not forget.

5. Sign in with your **Samsung account** when asked.
6. **Distributor certificate:** leave the privilege at **Public**. Your TV's DUID should already be listed (Certificate Manager reads it from the connected TV); if not, add it by hand.
7. Finish, and make sure the new profile is ticked as active.

> **YOU SHOULD SEE:** your profile (e.g. MyFamilyTV) in Certificate Manager with an author and a distributor certificate.

Keep the certificate files and password safe, forever, and out of git. *Every future update must be signed with the same author certificate. Lose it and you can only uninstall and start again. Back it up somewhere private. Got a second Samsung TV? Add its DUID to the distributor certificate too.*

30 MIN



PART 4 · THE APP *Point it, name it, send it*

9

Point the app at your station

The TV app is a little launcher that opens your station's TV page full screen. Everything for it lives in `apps/tv-samsung`. Your own settings go in a small file there called `.build-tv.local` (it stays on your computer, since git ignores it). Start it with your station's address, ending in `/tv`.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER

```
printf 'TV_SERVER=https://tv.your-domain.com/tv\n' > apps/tv-samsung/.build-tv.local
```

No domain? Use `https://saturday-morning.YOUR-SUBDOMAIN.workers.dev/tv`.

> **YOU SHOULD SEE:** the TV app when you open that exact address in a browser on your computer. If the browser can't open it, neither can the TV.

5 MIN



10

Make the app yours

Give the app its own id, so it can't clash with ours. Use your own **10 letters or numbers**, like `SatMorn042`, then `.SaturdayMornings`. The build puts it into the app's `config.xml` for you.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER

```
printf 'TIZEN_APP_ID=SatMorn042.SaturdayMornings\n' >> apps/tv-samsung/.build-
```

```
tv.local
```

Editing `apps/tv-samsung/config.xml` *yourself? Keep* `<name>` *as* `Saturday Mornings`*, because the build script renames the file by that exact name. And never add* `<tizen:allow-navigation>`*, or the TV shows a black screen.*

YOU SHOULD SEE: two lines in `apps/tv-samsung/.build-tv.local` when you open it, with your address and your id.

5 MIN

11

WOO!

Build, sign and install

One script does the lot. It connects to the TV if needed, builds the launcher, signs it with your certificate profile, packages it as a `.wgt` file (a TV app in a box) and installs it. Add your TV's address and your certificate profile to the same `.build-tv.local`, then run it.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER

```
printf 'TV_IP=192.168.1.50\nTIZEN_PROFILE=MyFamilyTV\n' >> apps/tv-samsung/.build-tv.local
apps/tv-samsung/build-tv.sh
```

YOU SHOULD SEE: the build and package steps scroll past, then a message that the package was installed. If Tizen asks for your certificate password, it's the one from step 8. You can also pass the address each time instead (`apps/tv-samsung/build-tv.sh 192.168.1.50`), or build the `.wgt` without a TV with `--package-only`.

10 MIN

PART 5 · ON THE AIR *Switch on, pair up, sit back*

12

First launch

On the TV, open **Apps** and find **Saturday Mornings** (new apps usually land at the end of the row). Open it. It checks your station is answering, then opens it full screen.

YOU SHOULD SEE: a pairing code and a QR code on the TV. Your station is nearly on the air.

2 MIN

ON AIR!

13 Link it from your phone

In the parent app on your phone, go to **Your TVs** › + **Link a TV** and type the code the TV shows.

> **YOU SHOULD SEE:** the TV switch to your kids' channels. From now on it checks for your changes every 30 seconds. Take a bow.

2 MIN

14 Updating the TV app later

Most updates only touch the station, and the TV picks them up by itself. Only when an update changes something in `apps/tv-samsung` (or you move your station) do you reinstall: TV on, developer mode still on, same certificate, same script.

TYPE THIS

```
apps/tv-samsung/build-tv.sh 192.168.1.50
```

> **YOU SHOULD SEE:** the same “installed” message as step 11. Moved your station? Link the TV again afterwards.

5 MIN

STUCK?

Black screen, or “Can't connect right now”	It's almost always the `TV_SERVER` address in `apps/tv-samsung/.build-tv.local`, with a typo, a missing `https://` or `/tv`, or a station that isn't up (check `/api/health`). Fix it, then run `apps/tv-samsung/build-tv.sh` again. Also check nobody added `<tizen:allow-navigation>` to `apps/tv-samsung/config.xml`.
`sdb connect` **fails or times out**	TV switched on? Same Wi-Fi? Is the IP in developer mode exactly your computer's current address? Home routers sometimes hand out new addresses, so check step 4 and step 6 again.
The TV stops accepting installs	Check Developer Mode is still on (step 5). It can switch off, for example after a TV reset. Turn it back on with the 1-2-3-4-5 trick and restart the TV.

Install refused: certificate or signature errors	The app must be signed with a **Samsung** certificate whose distributor half includes this TV's DUID. Check your profile is active in Certificate Manager, matches `TIZEN_PROFILE` in `apps/tv-samsung/.build-tv.local`, and the DUID matches. Updating? Use the same author certificate as the first install.
The script says `mv` can't find the `.wgt`	The `<name>` in `apps/tv-samsung/config.xml` changed. Put it back to `Saturday Mornings`, or edit the `mv` line in `build-tv.sh` to match. Keep spaces out of the final file name too, because the TV rejects them.
New icon doesn't show	A reinstall keeps the old icon. Uninstall the app on the TV first (Apps, highlight it, open its options, Delete), then run the script again.
`tizen: command not found`	Tizen Studio isn't in `~/tizen-studio`. Reinstall it there, or edit the `PATH` line at the top of `build-tv.sh`.
The pairing code won't link	Make sure the phone is signed in to the same station the TV points at. Type the code while it's still on screen.

Still stuck? Open an issue on the code's GitHub page with the command you ran and exactly what it said. Leave your certificate and passwords out of it.

That's a TV. *Grab the cereal, sit the kids down, and* **enjoy your Saturday mornings.** *Haven't built your station yet? Start there. Rather not do any of this? The home-delivery edition installs in three steps.*

